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Input Reduction for Scalable Reinforcement Learning in Quantum Computing Noise Models

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Abstract

In the NISQ (Noisy Intermediate-Scale Quantum) era, the mitigation and modeling of quantum noise are crucial for improving the reliability of quantum computations [9]. This thesis explores *input reduction* strategies in reinforcement learning (RL)-based quantum noise modeling, focusing on the case where the density matrix is restricted to its diagonal elements.

After introducing the fundamentals of quantum computing, noise channels, and the Proximal Policy Optimization (PPO) algorithm, the work evaluates whether such a reduced representation can retain the essential information for accurate noise characterization while lowering input dimensionality and improving scalability to larger systems.

Starting from the code for full density matrices [2] and adapting it to the purpose of this study, RL agents trained on *full* versus *diagonal-only* density matrices are compared, using fidelity and trace distance as evaluation metrics. Results show that the diagonal input preserves most global information relevant for fidelity optimization, while the trace distance highlights the expected loss of local correlations.

A second analysis investigates the role of measurement statistics by varying the number of *shots* in state tomography for 1- and 3-qubit systems. The reduced-input model maintains competitive performance even in low-shot regimes, with asymptotic fidelities compatible with $F_\infty \approx 1$ for both cases, while the trace distance does not vanish, indicating intrinsic limits in local information recovery. Defining the *critical shot number* as the minimum budget required to reach a fidelity of 0.95, we find a clear scaling with system size: for 1 qubit, ≈ 20 shots in the *full* setting and ≈ 10 in the *diagonal* setting; for 3 qubits, ≈ 100 shots in both the *full* and *diagonal* scenarios.

These findings support input reduction as a viable and scalable strategy for RL-based quantum noise modeling.

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Chapter 1

Quantum Computing

To understand quantum noise and its impact on quantum computation, we must first introduce the mathematical foundations of quantum mechanics and the basic structure of quantum circuits.

1.1 Mathematical Foundations

To properly describe a quantum computation model, it is essential to define the fundamental unit of information on which quantum computers are based:

Definition 1.1 (Quantum bit)**.** *The quantum bit (or qubit) is defined as the fundamental unit of quantum information. It is represented by a unit-norm vector belonging to a two-dimensional separable Hilbert space $\mathcal{H}$.*

In general, given a set of orthonormal vectors $\{|e_i\rangle\}_{i\geq 1}$ (whose existence is guaranteed by the separability of $\mathcal{H}$), each state can be associated with its components, thereby obtaining a *representation*. In particular, countable Hilbert spaces are isomorphic to the space of square-summable sequences ℓ^2 [8]:

$$|\psi\rangle \in \mathcal{H} \to \begin{bmatrix} c_1 \\ ... \\ c_\infty \end{bmatrix} \in \ell^2$$

In our case, since we require the qubit to be measurable in two distinct states, analogous to a classical bit, we define the following basis:

Definition 1.2 (Computational basis)**.** *The computational basis of the Hilbert space $\mathcal{H}$ is defined by the vectors:*

$$|0\rangle \to \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad |1\rangle \to \begin{bmatrix} 0 \\ 1 \end{bmatrix} \tag{1.1}$$

Each qubit, being an element of the Hilbert space $\mathcal{H}$, can be expressed as a linear combination of the computational basis vectors:

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle \to \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

where $\alpha, \beta \in \mathbb{C}$ and satisfy the normalization condition $|\alpha|^2 + |\beta|^2 = 1$.

A geometric representation of the qubit state can be particularly useful for understanding its operations and transformations. In this formulation, the qubit state takes the following form:

$$|\psi\rangle = \cos(\theta/2)\,|0\rangle + e^{i\phi}\sin(\theta/2)\,|1\rangle \rightarrow \begin{bmatrix} \cos(\theta/2) \\ e^{i\phi}\sin(\theta/2) \end{bmatrix}, \quad \theta \in [0, \pi]; \quad \phi \in [0, 2\pi]$$

where the term $e^{i\phi}$ represents the relative phase of the state. By interpreting θ and ϕ as polar coordinates, the qubit state can be visualized as a point in $\mathbb{R}^3$ on the *Bloch sphere*, as shown in Figure 1.1.

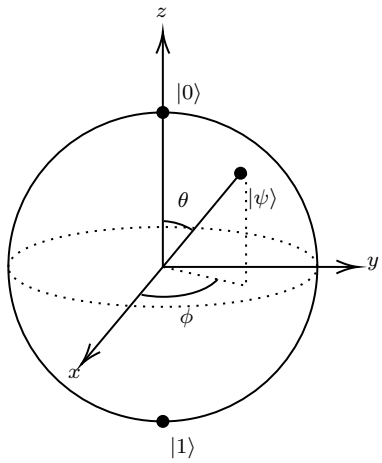


Figure 1.1. Bloch Sphere.

1.1.1 Entanglement

The main difference between a classical bit and a qubit lies in the latter's ability to exist in a superposition of states. This property introduces a fundamental challenge in quantum computing: extracting the information contained in the qubit through its measurement.

To perform a quantum computation, it is necessary to consider a system composed of n qubits. The overall state of such a system is described by a vector belonging to the Hilbert space obtained from the tensor product of the Hilbert spaces associated with the individual qubits:

$$\mathcal{H} = \mathcal{H}_1 \otimes \cdots \otimes \mathcal{H}_n.$$

Definition 1.3 (Factorized state)**.** *A state of qubits $|\psi\rangle \in \mathcal{H}$ is called factorized if it can be written as the tensor product of the states of the individual subsystems:*

$$|\psi\rangle = \bigotimes_{i=1}^{n} \left(\sum_{q_i} c_{q_i} |q_i\rangle \right). \tag{1.2}$$

with $q_i \in \mathbb{C}, \forall i \in \mathbb{N}; \quad \sum_i |q_i|^2 = 1$

Definition 1.4 (Entangled state)**.** *A state of qubits $|\psi\rangle \in \mathcal{H}$ is called entangled if it cannot be written in a factorized form, meaning that the system can only be described as a whole and not through the states of its individual subsystems:*

$$|\psi\rangle = \sum_{q_1,q_2,\ldots,q_n} c_{q_1,q_2,\ldots,q_n} |q_1 q_2 \ldots q_n\rangle . \tag{1.3}$$

with $q_i \in \mathbb{C}, \forall i \in \mathbb{N}; \quad \sum_i |q_i|^2 = 1$

In the case of a system of qubits, using the computational basis defined in 1.2, we have:

$$|\psi\rangle = \begin{cases} \bigotimes_{i=1}^{n} |q_i\rangle & \text{if the state is separable (factorized)} \\ \sum_{i=0}^{2^n-1} c_i |i\rangle & \text{if the state is entangled} \end{cases}$$

Here, n denotes the number of qubits in the system, while $|i\rangle$ represents the computational basis, encompassing all possible binary strings of length n. For an entangled state, each binary string $|i\rangle$ corresponds to a unique combination of qubit states, such as $|00\ldots00\rangle$, $|00\ldots01\rangle$, and so on, based on the value of i.

1.1.2 Unitary Operators and Quantum Gates

The state of a qubit can be altered by applying unitary operators, which serve as the mathematical foundation for *quantum gates*. These are essential components of quantum circuits, functioning similarly to classical logic gates in traditional computing.

Definition 1.5 (Unitary Operator)**.** *A unitary operator is an isomorphism $U : \mathcal{H}_1 \to \mathcal{H}_2$ between two Hilbert spaces for which*

$$\langle\varphi|\psi\rangle = \langle U\varphi|U\psi\rangle \quad \forall \varphi, \psi \in \mathcal{H}_1 \tag{1.4}$$

Alternatively, a unitary operator is one that satisfies:

$$U^\dagger U = UU^\dagger = I \tag{1.5}$$

where $U^\dagger$ denotes the adjoint of the operator U and I is the identity operator.

If we consider $\mathcal{H} = \mathbb{C}^n$, then every unitary operator can be represented by a unitary matrix $U \in U(n)$, which belongs to the group of $n \times n$ complex matrices satisfying the unitarity condition $U^\dagger U = UU^\dagger = I$.

The most general transformation of an n-qubit state is a unitary operation on the 2^n-dimensional Hilbert space, which enables the conversion of a product state of n qubits into any arbitrary entangled state, and vice versa.

Single-qubit gates

These operations act on individual qubit states. They are described by 2×2 unitary matrices with complex coefficients.

One of the most fundamental single-qubit gates is the X gate, which swaps the computational basis states $|0\rangle$ and $|1\rangle$ and so acts for this basis as a classical NOT gate[1]: using the computational basis it can be written as

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \tag{1.6}$$

Its action on the computational basis states is given by:

$$\begin{aligned} X|0\rangle &= \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = |1\rangle\,, \\ X|1\rangle &= \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle\,. \end{aligned}$$

Along with the X gate, two other fundamental single-qubit gates are the Y and Z gates. Together, they form the set of *Pauli matrices*:

$$X \equiv \sigma_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad Y \equiv \sigma_y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad Z \equiv \sigma_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \tag{1.7}$$

Another gate of fundamental importance is the Hadamard gate, which enables the creation of superpositions of qubits:

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \equiv \frac{1}{\sqrt{2}}(\sigma_x + \sigma_z) \tag{1.8}$$

$$\begin{aligned} H|0\rangle &= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \left(|0\rangle + |1\rangle\right) \equiv |+\rangle\,, \\ H|1\rangle &= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \frac{1}{\sqrt{2}} \left(|0\rangle - |1\rangle\right) \equiv |-\rangle\,. \end{aligned}$$

Here, the $|+\rangle$ and $|-\rangle$ states represent the *Hadamard basis*.

In general, the transformation of a qubit can be viewed as a rotation of the vector representing its state on the Bloch sphere. Specifically, finite rotations are derived as the limit of infinitely many applications of the generator of infinitesimal rotations, where $\vec{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$. The corresponding unitary operator is given by

$$R(\theta) = \cos(\theta/2)I - i\sin(\theta/2)\hat{n} \cdot \vec{\sigma}$$

[1] The hypothetical *universal* NOT operation would take an arbitrary qubit state $|\psi\rangle$ and replace it by its orthogonal complement $|\psi^{\perp}\rangle$. However, the u-NOT gate is impossible since it requires a *non-unitary transformation*. On the Bloch sphere, valid operations are rotations, but u-NOT would perform a reflection, which is not unitary, thus, it cannot be implemented. For more details see Chapter 21 [1].

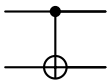
which leads to the unitary operator in polar coordinates [13]:

$$\exp\left(-\frac{i\theta}{2}\hat{n}\cdot\vec{\sigma}\right) = \begin{bmatrix} \cos(\theta/2) & -\sin(\theta/2)e^{-i\phi} \\ \sin(\theta/2)e^{i\phi} & \cos(\theta/2) \end{bmatrix} \quad \theta \in [0,\pi]; \quad \phi \in [0, 2\pi] \tag{1.9}$$

Multiple qubit Gates

These operations act on pairs of qubits. They enable the execution of Boolean logic operations and the generation of entangled states.

The most important of these is the *Controlled* NOT (CNOT) gate. This gate takes two qubits as input: one is the *control* qubit, and the other is the *target* qubit. If the control qubit is 0, the target qubit remains unchanged. However, if the control qubit is 1, the value of the target qubit is flipped.

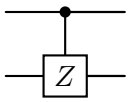
CNOT =  $= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$ (1.10)

$$\begin{aligned} \text{CNOT}\,|00\rangle &= |00\rangle \\ \text{CNOT}\,|01\rangle &= |01\rangle \\ \text{CNOT}\,|10\rangle &= |11\rangle \\ \text{CNOT}\,|11\rangle &= |10\rangle \end{aligned}$$

There are many other useful multiple qubit quantum gates: for instance, the SWAP gate exchanges the states of two qubits: $|ij\rangle \to |ji\rangle$.

SWAP = $= \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ (1.11)

Furthermore, the CZ gate applies a Z gate to the target qubit if the control qubit is in the state $|1\rangle$.

CZ =  $= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$ (1.12)

Similarly to the universality of the NAND gate in classical computing, any quantum circuit can be simulated to an arbitrary degree of accuracy using a combination of CNOT gates and single qubit rotations[2] as shown in Figure 1.2.

[2] A formal proof of this universality result is beyond the scope of this thesis and will not be presented here. For further details, see Chapter 4 Section 5.2 [6].

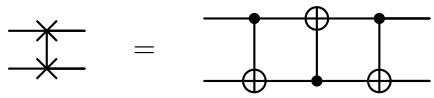


Figure 1.2. Gate decomposition into C-NOT gates of the SWAP gate.

Quantum gates can be used to generate entangled states. As an illustrative case, we consider the generation of a *Bell state* using a minimal circuit.

The Bell states form an orthonormal basis for the two-qubit Hilbert space and represent the simplest examples of maximally entangled states. They are given by:

$$
\begin{aligned}
|\Phi^+\rangle &= \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \\
|\Phi^-\rangle &= \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle) \\
|\Psi^+\rangle &= \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle) \\
|\Psi^-\rangle &= \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)
\end{aligned}
$$

Applying a Hadamard gate on the first qubit followed by a CNOT gate yields the state $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$, as shown in Figure 1.3.

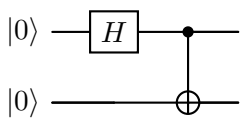


Figure 1.3. Circuit generating the Bell state $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

1.1.3 Measurements

While the Bloch sphere offers a useful geometric representation for a single qubit, it also suggests a paradox: since there are infinitely many points on the sphere, it might appear that a qubit could encode an infinite amount of information—for example, in the continuous parameters θ and ϕ. However, this intuition is misleading. Upon measurement, a qubit yields only a single classical bit—either 0 or 1—and the measurement process irreversibly alters the state of the system.

This phenomenon is formalized in quantum mechanics by two key postulates:

Definition 1.6 (Measurement Postulates)**.**

- ***Postulate I (Observables and spectra).*** *To every measurable quantity A, there corresponds a linear, self-adjoint operator $\hat{A}$ on a Hilbert space $\mathcal{H}$. The possible outcomes of a measurement of A are the elements of the spectrum of $\hat{A}$; in finite dimensions, these are simply its eigenvalues.*

- ***Postulate II (Measurement probabilities and collapse).*** *If the system is in a normalized state $|\psi\rangle$, and $\hat{A}$ has a spectral decomposition $\hat{A} = \sum_i a_i P_i$, then the probability of obtaining outcome a_i is given by:*

$$\mathbb{P}(a_i) = \langle\psi|P_i|\psi\rangle. \tag{1.13}$$

After the measurement, if outcome a is observed, the system collapses to the corresponding eigenstate:

$$|\psi\rangle \longrightarrow \frac{P_i|\psi\rangle}{\sqrt{\langle\psi|P_i|\psi\rangle}} \tag{1.14}$$

Where P_i is the projection operator onto the subspace corresponding to the eigenvalue a_i, satisfying $P_i^2 = P_i$ and $\sum_i P_i = I$, where I is the identity operator. These operators are also orthogonal, meaning $P_i P_j = 0$ for $i \neq j$

For example, consider a single qubit in the pure state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, where $\alpha, \beta \in \mathbb{C}$ and $|\alpha|^2 + |\beta|^2 = 1$. Measuring the qubit in the computational basis corresponds to measuring the observable associated with the Pauli-Z operator, which has eigenvalues $+1$ and -1, corresponding to the states $|0\rangle$ and $|1\rangle$, respectively.

According to the *Postulate II*, the probability of measuring the outcome associated with $|0\rangle$ is $|\alpha|^2$, and with $|1\rangle$ is $|\beta|^2$. Crucially, the act of measurement causes the state to collapse: if the outcome is 0, the state becomes $|0\rangle$ and vice versa, as a consequence of the collapse postulate.

Therefore, despite the fact that a qubit is described by continuous complex amplitudes α and β, a single measurement yields only *one classical bit of information.* To reconstruct the full quantum state, one would need to repeat the measurement many times on identically prepared qubits.

1.2 Decoherence and Quantum Noise

In the ideal quantum mechanical framework, an isolated system evolves unitarily according to the Schrödinger equation. However, real qubits are never perfectly isolated: interactions with the surrounding environment inevitably occur, leading to *decoherence* challenges for reliable quantum information processing.

An open quantum system is modeled as a subsystem of a larger composite system, where the environmental degrees of freedom are traced out. This loss of information induces non-unitary dynamics in the reduced system description.

To study noise effects, one typically models both the environment and its interaction with the system. These two elements determine how external disturbances affect the quantum state.

In most practical scenarios, exact modeling of the environment is infeasible. Still, with reasonable assumptions, one can build effective noise models that capture the dominant sources of error in real quantum devices.

1.2.1 Mathematical formalism

The behavior of a real qubit cannot, in general, be fully described by a pure state $|\psi\rangle$. Instead, one must use the density matrix formalism to account for statistical mixtures and interactions with the environment.

Definition 1.7 (Density matrix)**.** *A quantum state is described by a* density operator ρ*, which is a positive semi-definite, Hermitian operator with unit trace acting on a Hilbert space $\mathcal{H}$. For a pure state $|\psi\rangle$, the density operator is given by $\rho = |\psi\rangle\langle\psi|$. For a statistical mixture of states $\{|\psi_i\rangle\}$ with probabilities $\{p_i\}$, the density matrix is*

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i| \tag{1.15}$$

In the density matrix formalism, the description of quantum measurements also undergoes a generalization. While in the pure state formalism, a projective measurement associated with an observable A is described by a set of orthogonal projectors $\{P_i\}$ corresponding to the eigenstates of A, the outcome probabilities are computed as $\mathbb{P}(a_i) = \langle\psi| P_i |\psi\rangle$.

In the density matrix formalism, the probability of obtaining outcome a_i becomes

$$\mathbb{P}(a_i) = \mathrm{Tr}(P_i\rho),$$

which reduces to the previous expression when $|\psi\rangle$ is a pure state. This formulation remains valid even for mixed states, making it suitable for describing systems that are not in a well-defined pure state.

The density matrix formalism offers a suitable framework to describe the phenomenon of *decoherence*, whereby a quantum system loses its ability to exhibit coherent superpositions due to environmental coupling.

Definition 1.8 (Decoherence)**.** *Decoherence refers to the decay of the off-diagonal terms of the density matrix, expressed in a particular basis, as a result of entanglement with environmental degrees of freedom. This suppression of coherence eliminates interference effects and yields behavior that resembles classical stochastic dynamics.*

As an illustrative case, consider a qubit prepared in the coherent superposition:

$$|\psi\rangle = \frac{e^{i\theta_1}}{\sqrt{2}}|0\rangle + \frac{e^{i\theta_2}}{\sqrt{2}}|1\rangle .$$

The expectation value of an observable A in this state is computed as:

$$\langle A\rangle = \langle\psi| A |\psi\rangle = \frac{1}{2}\langle 0| A |0\rangle + \frac{1}{2}\langle 1| A |1\rangle + \mathrm{Re}[e^{i(\theta_1-\theta_2)}\langle 0| A |1\rangle].$$

The final term reflects the presence of quantum interference. However, when the system interacts with an external environment, even minimally, the relative phase $\theta = \theta_1 - \theta_2$ becomes unstable and undergoes random fluctuations. If the details of the environment are inaccessible, the correct approach is to average over all possible relative phase values:

$$\langle A\rangle = \frac{1}{2\pi}\int_0^{2\pi} d\theta\,\langle\psi| A |\psi\rangle = \frac{1}{2}\langle 0| A |0\rangle + \frac{1}{2}\langle 1| A |1\rangle ,$$

since

$$\int_0^{2\pi} d\theta \ \mathrm{Re}[e^{i\theta} \langle 0| A |1\rangle] = 0.$$

We now express the same situation using the density matrix formalism. The pure state $|\psi\rangle$ is associated with the operator

$$\rho = |\psi\rangle \langle\psi| = \frac{1}{2}\left(e^{i\theta_1}|0\rangle + e^{i\theta_2}|1\rangle\right)\left(e^{-i\theta_1}\langle 0| + e^{-i\theta_2}\langle 1|\right) =$$
$$= \frac{1}{2}\left(|0\rangle\langle 0| + |1\rangle\langle 1| + e^{i\theta}|0\rangle\langle 1| + e^{-i\theta}|1\rangle\langle 0|\right) = \begin{bmatrix} 1/2 & e^{i\theta}/2 \\ e^{-i\theta}/2 & 1/2 \end{bmatrix}.$$

Averaging over the random phase θ, we obtain a mixed state where coherence is fully suppressed:

$$\rho_{\mathrm{inc}} = \frac{1}{2}\left(|0\rangle\langle 0| + |1\rangle\langle 1|\right) = \begin{bmatrix} 1/2 & 0 \\ 0 & 1/2 \end{bmatrix}.$$

Note that this resulting density matrix no longer represents a pure state, since $\rho_{\mathrm{inc}}^2 \neq \rho_{\mathrm{inc}}$. Thus, decoherence has effectively transformed a coherent superposition into an incoherent probabilistic mixture.

To account explicitly for the role of the environment, consider the full quantum state after interaction:

$$|\Psi\rangle = \frac{e^{i\theta_1}}{\sqrt{2}}|0\rangle|e_1\rangle + \frac{e^{i\theta_2}}{\sqrt{2}}|1\rangle|e_2\rangle,$$

where $|e_1\rangle$ and $|e_2\rangle$ are environment states correlated with the system states $|0\rangle$ and $|1\rangle$, respectively. By tracing out the environment from the total state operator $\rho_{\mathrm{full}} = |\Psi\rangle\langle\Psi|$, we obtain the reduced density matrix of the system:

$$\rho = \mathrm{Tr}_{\mathrm{env}}(\rho_{\mathrm{full}}) = \begin{bmatrix} \frac{1+|\langle e_2|e_1\rangle|^2}{2} & e^{i\theta}\langle e_2|e_1\rangle \\ e^{-i\theta}\langle e_1|e_2\rangle & \frac{1+|\langle e_2|e_1\rangle|^2}{2} \end{bmatrix}.$$

Assuming normalized environment states ($\langle e_1|e_1\rangle = \langle e_2|e_2\rangle = 1$), the coherence between $|0\rangle$ and $|1\rangle$ is preserved only if the environment states remain indistinguishable. If $|e_1\rangle$ and $|e_2\rangle$ become orthogonal, the term $\langle e_1|e_2\rangle \to 0$, and the system density matrix collapses into the incoherent form:

$$\rho_{\mathrm{inc}} = \begin{bmatrix} 1/2 & 0 \\ 0 & 1/2 \end{bmatrix}.$$

1.2.2 Kraus Representation

In closed quantum systems, the time evolution of a state is governed by a unitary operator U, such that a density matrix ρ evolves as $\rho \to U\rho U^\dagger$. However, when the system interacts with an external environment, this description no longer suffices. The system is no longer isolated, and the influence of the surrounding degrees of freedom must be taken into account.

Let us assume that the system and the environment initially form a product state in the Hilbert space $\mathcal{H}_{\mathrm{sys}} \otimes \mathcal{H}_{\mathrm{env}}$. Suppose the system starts in a general

mixed state ρ_{sys}, and the environment is initialized in a pure state ρ_{env}, which we can take as $\rho_{\text{env}} = |0\rangle_{\text{env}} \langle 0|_{\text{env}}$ without loss of generality. The joint system evolves unitarily under a global operator U, and the reduced state of the system is obtained by tracing over the environment:

$$\rho' = \mathcal{E}(\rho) = \text{Tr}_{\text{env}} \left(U(\rho_{\text{sys}} \otimes \rho_{\text{env}}) U^\dagger \right),$$

where $\mathcal{E}(\rho)$ defines a *linear map*. As shown in Figure 1.4, if the evolution operator U does not couple the system and the environment, the evolution reduces to the familiar unitary form $\rho' = \tilde{U}\rho\tilde{U}^\dagger$, where $\tilde{U}$ acts solely on the system.

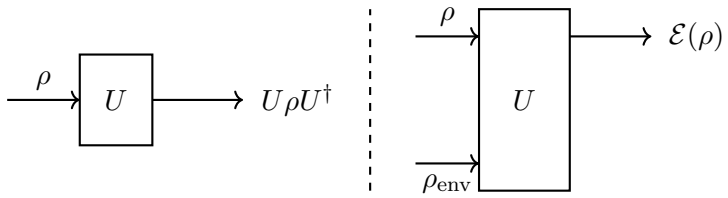


Figure 1.4. Time evolution in closed (left) and open (right) quantum systems. (Figure adapted from [6]).

Now consider an orthonormal basis $\{|e_k\rangle\}$ for the environment's Hilbert space $\mathcal{H}_{\text{env}}$. Then the reduced system state becomes:

$$\rho' = \sum_k \langle e_k| U |0\rangle_{\text{env}} \rho_{\text{sys}} \langle 0|_{\text{env}} U^\dagger |e_k\rangle .$$

Defining the operators $E_k = \langle e_k| U |0\rangle_{\text{env}}$, we arrive at the *Kraus representation* of a *quantum channel*:

Definition 1.9 (Quantum Channel)**.** *A quantum channel is a linear map $\mathcal{E} : \mathcal{B}(\mathcal{H}) \to \mathcal{B}(\mathcal{H})$ that satisfies:*

- ***Complete positivity**: $\mathcal{E}(\rho) \geq 0$ for all $\rho \geq 0$, even when extended to larger systems.*
- ***Trace preservation**: $\text{Tr}[\mathcal{E}(\rho)] = \text{Tr}[\rho]$, ensuring that the total probability is conserved.*

Every Completely-Positive Trace-Preserving (CPTP) map admits a Kraus decomposition:

$$\mathcal{E}(\rho) = \sum_k E_k \rho E_k^\dagger \tag{1.16}$$

where $\{E_k\}$ are the Kraus operators satisfying $\sum_k E_k^\dagger E_k = I$.

Each Kraus operator E_k corresponds to a possible outcome of the system's interaction with the environment. Since we do not measure the environment, we sum over all possible outcomes, leading to the mixed final state. The Kraus representation is not unique[3] — different sets of Kraus operators can represent the same channel —but they all yield the same final state $\mathcal{E}(\rho)$.

[3]For further details, see Theorem 8.2 in [6]

1.2.3 Noise Channels

The Kraus representation provides a versatile framework to describe the most common types of quantum noise, each corresponding to a specific interaction between a quantum system and its environment.

Bit-Flip Channel: the bit-flip channel represents the quantum analogue of a *classical bit error*, where a qubit $|0\rangle$ is flipped to $|1\rangle$ with probability p and vice versa with probability $1-p$. This process can be modeled using the following Kraus operators:

$$E_0 = \sqrt{1-p}\, I, \quad E_1 = \sqrt{p}\, X,$$

where p is the probability of a bit-flip error occurring, and X is the Pauli-X (NOT) operator. The action of the channel on a state ρ is given by:

$$\mathcal{E}_{\text{bit-flip}}(\rho) = (1-p)\rho + p\, X\rho X.$$

Phase-Flip Channel: the phase-flip channel introduces a *phase error*, flipping the sign of the $|1\rangle$ component of a qubit state while leaving $|0\rangle$ unchanged. It is described by the Kraus operators:

$$E_0 = \sqrt{1-p}\, I, \quad E_1 = \sqrt{p}\, Z,$$

with Z being the Pauli-Z operator. The resulting transformation of the state is:

$$\mathcal{E}_{\text{phase-flip}}(\rho) = (1-p)\rho + p\, Z\rho Z.$$

Geometrically, the Bloch vector is projected along the z axis, and the x and y components of the Bloch vector are lost.

Depolarizing Channel: the depolarizing channel models a process in which the qubit *loses information* about its state with some probability p, becoming the *completely mixed state*. This noise acts isotropically and is described by the map:

$$\mathcal{E}_{\text{depol}}(\rho) = (1-p)\rho + \frac{p}{3}(X\rho X + Y\rho Y + Z\rho Z),$$

where X, Y, Z are the Pauli matrices. The corresponding Kraus operators are:

$$E_0 = \sqrt{1-p}\, I, \quad E_1 = \sqrt{\frac{p}{3}}\, X, \quad E_2 = \sqrt{\frac{p}{3}}\, Y, \quad E_3 = \sqrt{\frac{p}{3}}\, Z.$$

Amplitude Damping Channel: the amplitude damping channel captures the effect of *energy dissipation*, such as spontaneous emission in a two-level atom. It represents a transition from the excited state $|1\rangle$ to the ground state $|0\rangle$, with probability γ. The Kraus operators are:

$$E_0 = \begin{bmatrix} 1 & 0 \\ 0 & \sqrt{1-\gamma} \end{bmatrix}, \quad E_1 = \begin{bmatrix} 0 & \sqrt{\gamma} \\ 0 & 0 \end{bmatrix}.$$

The evolution under this channel is:

$$\mathcal{E}_{\text{amp-damp}}(\rho) = E_0 \rho E_0^\dagger + E_1 \rho E_1^\dagger.$$

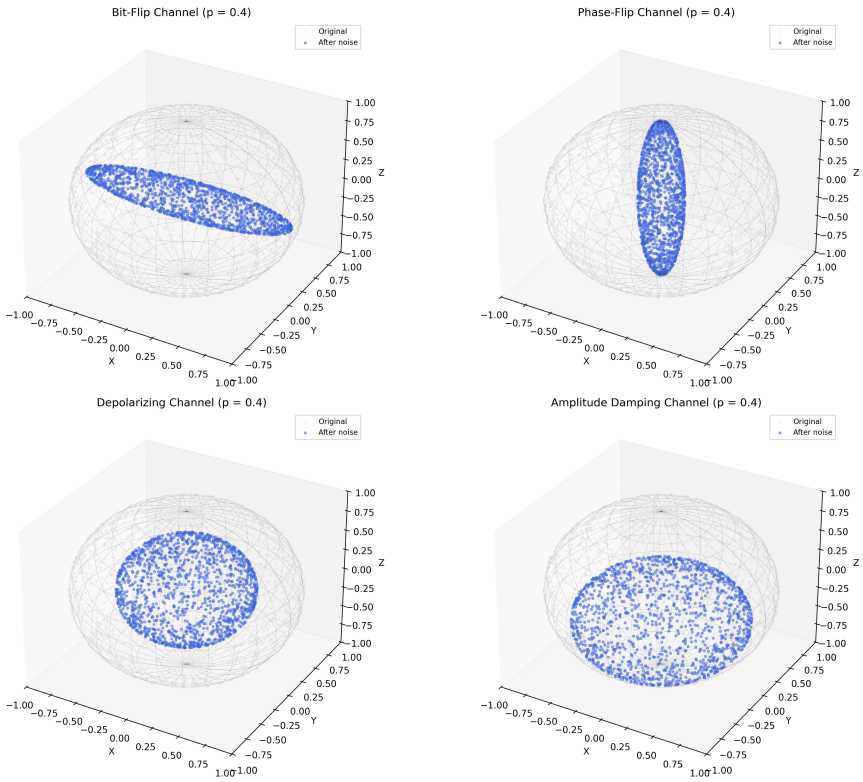


Figure 1.5. Effect of different quantum noise channels on the Bloch sphere. The initial pure quantum states, uniformly distributed on the surface of the sphere, are shown in light gray. The same states after the application of a noise channel with probability $p = 0.4$ are shown in blue.The figure was generated by the author using Python and the code is available at the following link.

As shown in Figure 1.5, each of these noise channels can be visualized as a distortion of the qubit's state on the Bloch sphere, with different models shrinking, flipping, or biasing the state vector.

Having introduced the mathematical framework of quantum computation and the main sources of noise affecting real quantum devices, we now shift our focus to tools that can help us learn and model such noise. In particular, the next chapter presents the fundamentals of Reinforcement Learning, a machine learning paradigm well-suited for decision-making tasks, and explores how it can be applied to characterize quantum noise effectively.

Chapter 2

Reinforcement Learning

2.1 Key Concepts

Reinforcement learning refers to a class of methods where an agent learns by interacting with an environment, rather than from labeled data.

Unlike *supervised learning*, which relies on input-output pairs to generalize to unseen examples, reinforcement learning is suited for *dynamic environments* where explicit supervision is impractical. It also differs from *unsupervised learning*, which finds patterns in unlabeled data but lacks feedback-driven adaptation. Thus, reinforcement learning stands as a distinct and complementary *paradigm* in machine learning.

A Reinforcement Learning algorithm is mathematically described as a Markov Decision Process.

Definition 2.1 (Markov Decision Process)**.** *A Markov Decision Process is a 5-tuple* (S, A, P, R, γ)*, where:*

- *S is the set of all the possible environmental states, also called the state space. It can be either discrete or continuous;*
- *A is the set of actions, also called the action space;*
- *$P(s'|s, a)$ is the probability that action a in the state s at time t will lead to the state s' at the time $t' = t + 1$;*
- *R(s,a) denotes the immediate reward received after transitioning from state s to s' due to action a;*
- *γ is the discount factor which balances the importance of immediate rewards against future ones.*

Solving a Markov Decision Process means finding a *policy function* $\pi(s) : S \to A$ that maximizes the expected average sum of rewards: it can be deterministic or stochastic, and can return the action to be taken in state s.

In a reinforcement learning model, the policy is implemented by a *Neural Network*, which is trained by performing different episodes of agent-environment interaction. At the end of each episode (or batch of episodes), the average episode reward is calculated and used to update the weights using the *backpropagation algorithm*

2.2 Modern RL Algorithms

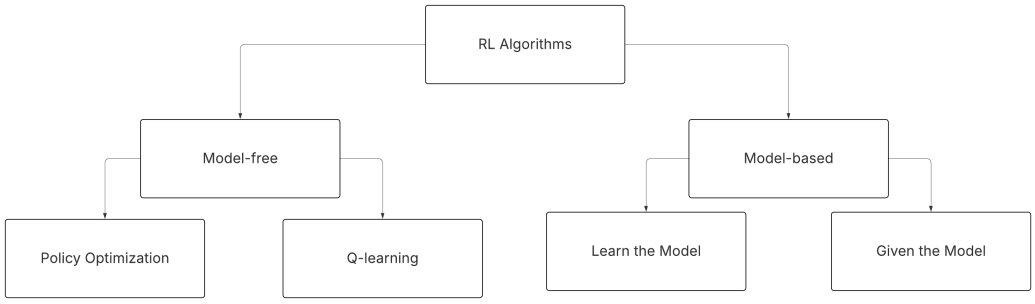


Figure 2.1. Algorithms in modern Reinforcement Learning. (Figure adapted from [7]).

A first key distinction in reinforcement learning methods concerns the availability of the *environment's dynamics*, and so of a function (*model*) which predicts state transitions and rewards. In *model-based* reinforcement learning, the agent either has full access to or learns an explicit model of the transition dynamics that govern how actions affect the environment. Essentially the agent can plan by thinking ahead, seeing what would happen for a range of possible choices, and explicitly deciding between its options.

In contrast, *model-free* reinforcement learning assumes no prior knowledge of the environment's dynamics. Instead, the agent learns directly from interaction with the environment.

Another pivotal design choice in reinforcement learning is *what the agent decides to estimate*. Depending on this choice, an algorithm may fall into a different family. Figure 2.1 summarizes this taxonomy.

- **Policy Optimization Methods** (Model-free): they estimate *Policy functions* $\pi_\theta(s_t)$ which depend on a set of parameters θ, and can be distinguished in:
  - *Stochastic policy* $\pi(a \mid s)$, which returns a probability distribution over actions.
  - *Deterministic policy* $\mu(s)$, which returns a single action for each state.

  These frameworks can also give an estimate of the *State–value functions* – the expected return starting from state s under policy π:

  $$V^\pi(s) \;=\; \mathbb{E}\,\big[G_t \,\big|\, S_t = s\big].$$

  Policy optimization methods optimize the parameters θ of $\pi_\theta(a \mid s)$ either by *gradient ascent* on $J(\pi_\theta)$ or by maximizing local approximations. This optimization is often performed *on-policy*, i.e. the agent updates values based on the action actually taken by the current policy.

- **Q-Learning Methods** (Model-free): they estimate *Action–value functions (Q–functions)* – the expected return $J(\pi) = \mathbb{E}[R(\tau)]$ (cumulative reward over

a sequence of states and action τ) obtained by taking action a in state s and thereafter following the optimal policy $\pi^* = \arg\max_{\pi} J(\pi)$:

$$Q^{\pi}(s,a) \;=\; \mathbb{E}\left[G_t \,|\, S_t = s,\; A_t = a\right].$$

This optimization is typically carried out *off-policy*, meaning that each update can leverage data collected at any stage of training, regardless of the exploration strategy the agent was following when that data was generated.

- **Model-based methods** can estimate the *Environment models* – explicit estimates of the dynamics $P(s' \mid s,a)$ and/or the reward function $R(s,a)$, enabling planning or model–based updates. There can also be distinguished two different types of these algorithms:
  - **Learn the model** – the agent builds its own dynamics approximation
  - **Given the model** – the dynamics are supplied and the agent plans inside them

2.2.1 Proximal Policy Optimization

Proximal Policy Optimization (PPO) is a *policy gradient* algorithm that updates the policy parameters to maximize the expected return via the gradient $\nabla_\theta J(\theta)$. Rather than optimizing $J(\theta)$ directly, PPO maximizes a *surrogate objective* that provides a conservative estimate of the performance improvement. PPO belongs to the *actor-critic* family: the agent is split into an *actor*, which proposes actions, and a *critic*, which estimates their value.

This method improves training stability through a *clipping* mechanism, which limits large and potentially harmful updates.

PPO is used in this thesis for its robustness and its natural handling of *continuous action spaces* [12], which is crucial in quantum noise modeling, where the learned parameters (e.g., depolarization rates, dephasing strengths) are real-valued.

The gradient of the expected return can be written as:

$$\nabla_\theta J(\theta) \;=\; \mathbb{E}_t\big[\,\nabla_\theta \log \pi_\theta(a_t \,|\, s_t)\; A_t^{\pi_\theta}\big]$$

where π_θ is a stochastic policy and $A_t^{\pi_\theta}(s,a) = Q^{\pi_\theta}(s,a) - V^{\pi_\theta}(s)$ is an estimator of the *advantage function*, which measures "how much better than average" action a is in state s.

Vanilla policy-gradient methods may take steps that are too large, making the new policy deviate dramatically from the old one and destabilizing learning. PPO addresses this with a clipped surrogate loss

$$\mathcal{L}_{\text{clip}}(\theta) = \mathbb{E}_t\Big[\min\big(r_t(\theta)\, A_t^{\pi_\theta},\; \text{clip}(r_t(\theta),\, 1\pm\varepsilon)\, A_t^{\pi_\theta}\big)\Big] \tag{2.1}$$

where $r_t(\theta) = \dfrac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)}$ is the surrogate objective, i.e. the *probability ratio* between the new policy and the policy that generated the data and $\varepsilon \in [0.1, 0.3]$ is the *clip parameter*. The clip function truncates $r_t(\theta)$ to the values $[1-\varepsilon,\, 1+\varepsilon]$.

To better understand this loss, let's analyze the different cases, as shown in Figure 2.2:

- If the advantage is positive, i.e. $A_t^{\pi_\theta}(s,a) > 0$, increasing the action probability $\pi_\theta(a \mid s)$ improves the objective. However, PPO limits this gain via

$$\mathcal{L} = \min\left(\frac{\pi_\theta(a \mid s)}{\pi_{\theta_k}(a \mid s)},\ 1+\epsilon\right) \cdot A_t^{\pi_\theta}(s,a).$$

  Once the ratio exceeds $1+\epsilon$, the improvement is capped and prevents the policy from drifting too far from the previous one.

- If the advantage is negative, decreasing $\pi_\theta(a \mid s)$ helps. The surrogate becomes

$$\mathcal{L} = \max\left(\frac{\pi_\theta(a \mid s)}{\pi_{\theta_k}(a \mid s)},\ 1-\epsilon\right) \cdot A_t^{\pi_\theta}(s,a).$$

  PPO limits how much this decrease can affect the update. Once the ratio falls below $1-\epsilon$, further reduction has no effect.

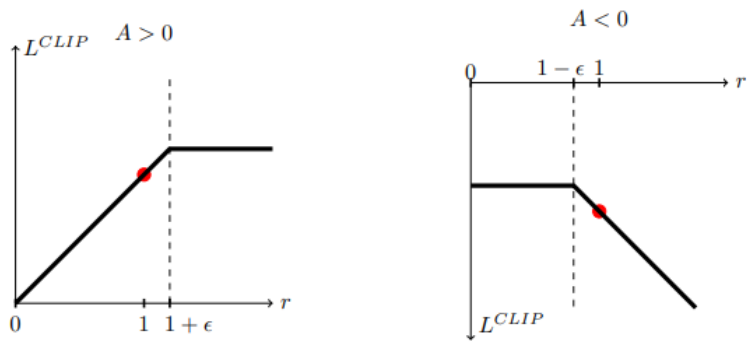


Figure 2.2. The Clipping term with respect to r_t. The red circle on each plot shows the starting point for the optimization (= 1). [12]

In practice, the exact advantage—and thus the true value function—is not accessible. When using a neural network that shares parameters between the policy and the value estimator, it is necessary to define a loss function that jointly accounts for both. This is done by combining the policy surrogate objective with a value function error term.

Putting all components together, we obtain the following overall objective:

$$\mathcal{L}(\theta,\phi) = \underbrace{\mathcal{L}_{\text{clip}}(\theta)}_{\text{policy}} + c_v\ \underbrace{\mathbb{E}_t\big[(V_\phi(S_t) - \hat{V}_t^{\text{target}})^2\big]}_{\text{value}} - c_s\ \underbrace{\mathbb{E}_t\big[\mathcal{H}(\pi_\theta(S_t))\big]}_{\text{entropy}} \tag{2.2}$$

where $\hat{V}_t^{\text{target}} = A_t^{\pi_\theta} + V_\phi(S_t)$, $\mathcal{H}(\,\cdot\,)$ is the Shannon entropy, that maximized it encourages exploration, and c_v, c_s are positive scalars weighting the two auxiliary terms. The overall optimization problem then is $\theta^\star, \phi^\star = \arg\max_{\theta,\phi}\ \mathcal{L}(\theta,\phi)$.

A popular estimator for the advantage function is the Generalized Advantage Estimation (GAE):

$$A_t^{\pi_\theta} = \sum_{l=0}^{T-1-t} (\gamma\lambda)^l \left[r_{t+l} + \gamma V_\phi(s_{t+l+1}) - V_\phi(s_{t+l})\right] \tag{2.3}$$

where $0 < \gamma < 1$ is the discount factor, T is the *rollout horizon*, i.e., the length of the trajectory of states and actions, and λ an hyperparameter that balances bias and variance, i.e., how much the agent should trust the critic's value estimates compared to the actual rewards observed from the environment.

If the action taken leads to rewards greater than what the critic expected, the advantage will be positive, hence the agent is encouraged to favor that action.

On the other hand if it leads to lower-than-expected rewards, the advantage is negative and so the agent is discouraged from repeating that action.

The PPO training loop is summarized in the pseudocode below.

Algorithm 1: Proximal Policy Optimization (PPO-clip)

Input : Initial policy π_θ, value network V_ϕ, clip parameter ε, GAE factor λ, rollout horizon T, number of epochs E, minibatch size B

Output: Updated parameters θ, ϕ

for *each iteration* **do**
- Run policy π_θ for T time steps and collect trajectories: $\mathcal{D} = \{(s_t, a_t, r_t)\}_{t=0}^{T-1}$
- Use V_ϕ and GAE(λ) to estimate: $\hat{R}_t$, $A_t^{\pi_\theta}$
- Store reference policy: $\theta_{\text{old}} := \theta$
- **for** *epoch* $= 1$ *to* E **do**
  - Shuffle $\mathcal{D}$ and divide into mini-batches of size B
  - **foreach** *minibatch* $\mathcal{B}$ **do**
    - Compute:
      - $r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}$
      - $\mathcal{L}_{\text{clip}}(\theta)$ from clipped surrogate
      - Value loss: $(V_\phi(s_t) - \hat{R}_t)^2$
      - Entropy bonus: $\mathcal{H}[\pi_\theta(\cdot \mid s_t)]$
    - Perform gradient ascent with Adam on total loss:

$$\mathcal{L} = \mathcal{L}_{\text{clip}}(\theta) - c_v \cdot \mathcal{L}_{\text{value}}(\phi) + c_s \cdot \mathbb{E}_t\big[\mathcal{H}\big(\pi_\theta(S_t)\big)\big]$$

Having introduced the reinforcement learning framework and the PPO algorithm, we now turn to its application in the context of quantum noise modeling. Specifically, we investigate whether feeding only the diagonal elements of the density matrix is sufficient for the agent to learn an effective noise model. It is also examined how the number of measurement shots impacts training stability for both configurations.

Chapter 3

Quantum Noise and PPO

3.1 Why Reinforcement Learning?

A variety of techniques have been developed to characterise noise on quantum processors, each balancing information content against experimental and classical cost:

- **Analytical master–equation models.** Markovian and non-Markovian master equations (e. g. Lindblad) yield a microscopic description of decoherence and a description of *system–environment* interactions, but require detailed knowledge of the environment that is rarely available on present-day hardware [3];
- **Quantum Process Tomography (QPT).** By preparing a tomographic set of input states and measuring in multiple bases, QPT reconstructs the full process matrix. The number of settings and the classical post-processing both scale as 4^{2n}, so the method is practical only for one– and two-qubit gates [14];
- **Randomized Benchmarking (RB).** RB applies random Clifford sequences[1] of increasing length followed by an inverting gate that ideally returns the system to its initial state; the survival probability of measuring the initial state after performing the sequence decays exponentially, giving the average gate fidelity[2]. RB is proven to be scalable but they do not allow for a detailed and localized characterization of individual noise sources [5];
- **Bayesian online methods.** The noise map is treated as a random variable: Sequential Monte Carlo updates a posterior after each batch of data, offering rigorous uncertainty quantification. The particle count and likelihood evaluations grow rapidly with circuit width, making the approach computationally expensive beyond a few qubits [4].

Reinforcement Learning inherits the strengths of the previous methods: it needs no explicit noise model, unlike master-equation or Bayesian schemes, and, once

[1] Clifford gates map Pauli operators onto Pauli operators under conjugation and are generated by H, S, and CNOT.

[2] Fidelity measures state or channel similarity. For pure states $|\psi\rangle, |\phi\rangle$, $F = |\langle\psi|\phi\rangle|^2$; for mixed states it generalises to $F(\rho,\sigma) = \left(\mathrm{Tr}\sqrt{\sqrt{\rho}\sigma\sqrt{\rho}}\right)^2$.

trained, yields noise predictions for unseen circuits in a single forward pass, unlike QPT and Bayesian filters.

Here we train a *Proximal Policy Optimization* agent to insert virtual noise channels along a circuit timeline, maximizing a reward that matches experimental outcomes. The learned model attains high fidelity and low trace-distance[3] errors on small multi-qubit circuits [2], providing a scalable, data-driven alternative.

3.2 Dataset and Training Procedure

The reinforcement-learning framework proposed in [2] treats an entire quantum circuit as a three-dimensional tensor —the *Quantum Circuit Representation* (QCR) — having the shape of [*qubits, depth, encoding*]: the first dimension saves the circuit's qubits, the second the parallel-executable gate layers, and the third encodes all gates and noise channels acting on each qubit at a given moment.

For the native gate set $\{R_x(\theta), R_z(\theta), \mathrm{CZ}\}$, the first slots encode the presence of R_x, R_z, and CZ gates, the fourth stores the normalized rotation angle, and the remaining four hold the parameters for depolarizing, amplitude-damping, and coherent R_z and R_x errors.

Because circuit depths vary, a sliding "kernel" window of size $k = 3$ is introduced so that, at moment m, the agent observes only the k neighbouring moments, focusing on local spatio-temporal correlations between gates and noise.

During training, each episode begins with a randomly drawn noiseless circuit from the training pool. At every moment the agent consults the current QCR window and outputs a matrix whose rows correspond to qubits and whose four columns [4] carry the continuous parameters of the allowable noise channels. These channels are then inserted into the circuit, updating the QCR.

When the final moment is processed the environment simulates the full noisy circuit, producing a density matrix ρ_{agent}; the agent then receives a reward function that is given by:

$$R = \frac{1}{\alpha\,\mathrm{TD}(\rho_{\text{agent}}, \rho_{\text{true}})^2 + \varepsilon}, \quad \alpha = 20,\ \varepsilon = 1 \times 10^{-15} \tag{3.1}$$

The policy mapping a QCR window to noise parameters uses a 2-D convolutional layer (16 filters for 1 qubit, 32 for 3 qubits) to extract spatio-temporal features, followed by separate 256-neuron MLPs as actor and critic in the PPO scheme implemented with `Stable-Baselines3` [10] and `PyTorch`.

Datasets are generated from circuits using the native gate set $\{R_x(\pi/2),\ R_z(\theta),\ \mathrm{CZ}\}$. For one qubit, 20 random Clifford circuits of depth 10 avoid overfitting [2]. For three qubits, the pool combines fixed-depth random circuits with independent gate sampling and exact decompositions of random Clifford unitaries. Ground-truth density matrices are computed analytically, using the noise model in Table 3.1.

[3]The trace distance $\mathrm{TD}(\rho, \sigma) = \frac{1}{2}\|\rho - \sigma\|_1$ equals 0 for identical states and 1 for perfectly distinguishable states.

[4]This work considers only four possible noise channels: depolarizing, amplitude damping, $R_x(\theta)$ and $R_z(\theta)$.

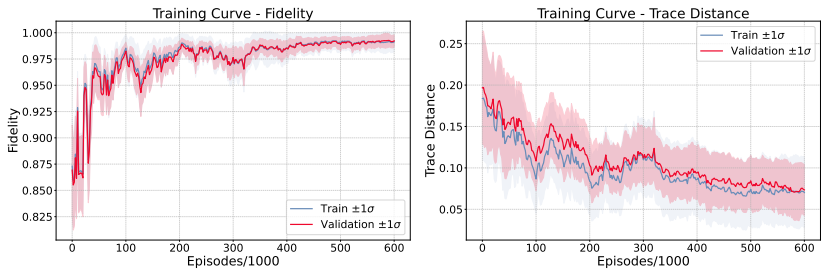
Qubits	Gate	λ	γ	θ_x	θ_z
1	$R_x(\theta)$	0	0.03	$0.04{\cdot}\theta$	0
	$R_z(\theta)$	0.02	0	0	$0.02{\cdot}\theta$
3	$R_x(\theta)$	0	0.01	$0.015{\cdot}\theta$	0
	$R_z(\theta)$	0.015	0	0	$0.02{\cdot}\theta$
	CZ	0.015	0.01	0	0

Table 3.1. Noise model parameters used in simulations for one- and three-qubit datasets: λ (depolarising), γ (amplitude damping), and θ_x, θ_z (coherent over-rotations) applied after each gate.

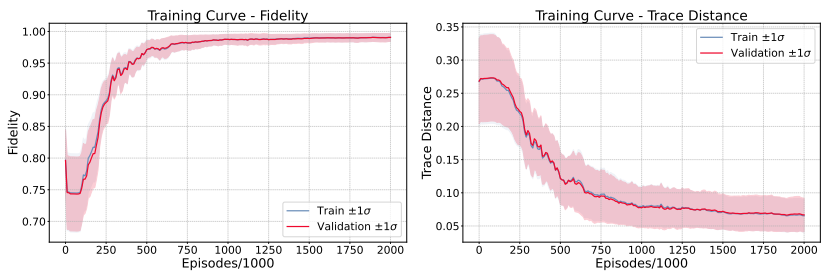
The one-qubit dataset includes 100 Clifford circuits (depth 10), split 80/20 for training and test, plus 100 evaluation circuits (depth 15). The three-qubit dataset has 800 mixed circuits (depth 20) split 80/20, with 200 unseen evaluation circuits of depth 15 for validation.

3.3 Full Density Matrix

The results obtained[5] training the PPO agent with the full density matrices for both types of circuits are consistent with the ones shown in [2] and are shown in Figure 3.1:



(a) 1-qubit training.



(b) 3-qubit training.

Figure 3.1. Evolution of fidelity and trace distance during training. Shaded areas represent the standard deviation across circuits, highlighting the variability of the agent's performance.

[5]The full implementation, including all scripts for dataset generation, model training, and evaluation, is available at the following link: Google Drive Repository.

No signs of overfitting can be observed during the training phase. The best-performing model was obtained at episode 5.9×10^5 for one-qubit circuits and at episode 1.89×10^6 for three-qubit circuits, achieving the results for the train, validation and test sets shown in Table 3.2:

Qubits	Set	Fidelity	Trace Dist.	Num. Err.	Exec. Time
1	Train	0.992 ± 0.008	0.070 ± 0.043	0.034	0:40:52
	Validation	0.993 ± 0.006	0.071 ± 0.034		
	Test	0.989 ± 0.013	0.086 ± 0.049		
3	Train	0.991 ± 0.006	0.065 ± 0.025	0.047	6:43:44
	Validation	0.991 ± 0.006	0.066 ± 0.025		
	Test	0.988 ± 0.006	0.076 ± 0.022		

Table 3.2. RL-agent performance on one- and three-qubit datasets. Uncertainties are one standard deviation across circuits; *Numerical Error* is the trace-distance uncertainty from sampling, readout and depolarisation, and *Execution Time* is the total training + evaluation runtime.

3.4 Diagonal Elements

In the diagonal–only setting the evaluation metrics were adapted accordingly. For two probability vectors p, q (the density-matrix diagonals) the trace distance and fidelity become:

$$\mathrm{TD}(\rho,\sigma) = \frac{1}{2}\,\|\rho-\sigma\|_1 \xrightarrow{\text{diagonal}} \mathrm{TD}(p,q) = \frac{1}{2}\sum_i |p_i - q_i|$$

$$F(\rho,\sigma) = \left(\mathrm{Tr}\big[\sqrt{\sqrt{\rho}\,\sigma\sqrt{\rho}}\big]\right)^2 \xrightarrow{\text{diagonal}} F(p,q)\Big(\sum_i \sqrt{p_i q_i}\Big)^2$$

Empirically it was found that, in the diagonal setting, the most effective and stable form of the reward was:

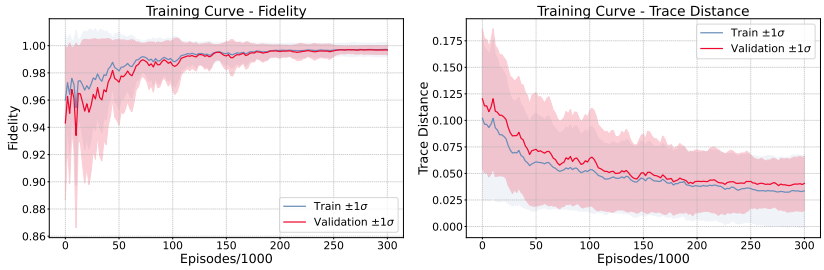
$$R(p,q) \;=\; -\log\left(\alpha\,\mathrm{TD}_{\mathrm{diag}} + \varepsilon\right), \quad \alpha = 0.5,\ \varepsilon = 1\times 10^{-15} \tag{3.2}$$

The difference in reward functions stems from the distinct error scales in the two settings. With full matrices, residual trace distances are very small (TD $<$ 0.05), so a sharp reward $R \propto 1/\mathrm{TD}^2$ amplifies tiny improvements. With diagonal inputs, missing off-diagonal terms set a higher error floor; a logarithmic reward keeps gradients stable and avoids numerical blow-ups.[6]

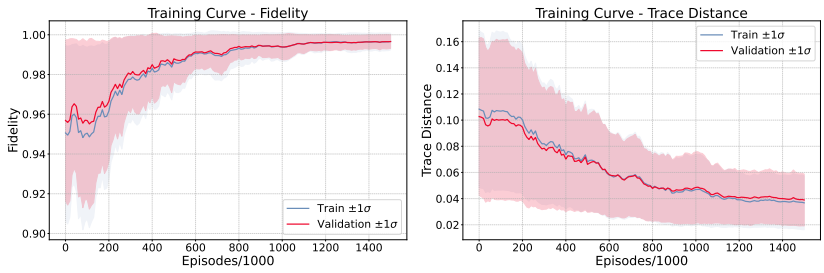
Results are shown in Figure 3.2:

No signs of overfitting were observed during training. As expected, the best-performing model was reached earlier with diagonal inputs, at episode $2.76 \times 10^5 \ll 5.9 \times 10^5$ for one-qubit circuits and $1.25 \times 10^6 < 1.89 \times 10^6$ for three-qubit circuits, achieving the train, validation, and test results reported in Table 3.3.

[6]Using $1/\mathrm{TD}^2$ instead of the logarithmic form for 1 qubit circuits produces a training-signal dynamic range of $\log_{10}(\max/\min) = 4.886$ and leads to a final validation fidelity of 0.943 ± 0.061. With the logarithmic reward, the dynamic range shrinks to 0.151 and the final validation fidelity is 0.997 ± 0.003.



(a) 1-qubit training.



(b) 3-qubit training.

Figure 3.2. Evolution of fidelity and trace distance during training. Shaded areas represent the standard deviation across circuits, highlighting the variability of the agent's performance.

Qubits	Set	Fidelity	Trace Dist.	Num. Err.	Exec. Time
1	Train	0.997 ± 0.005	0.032 ± 0.032	0.034	0:20:58
	Validation	0.997 ± 0.003	0.039 ± 0.023		
	Test	0.984 ± 0.021	0.098 ± 0.053		
3	Train	0.997 ± 0.004	0.037 ± 0.021	0.045	4:07:44
	Validation	0.997 ± 0.003	0.039 ± 0.020		
	Test	0.983 ± 0.008	0.094 ± 0.027		

Table 3.3. RL-agent performance on one- and three-qubit datasets trained only on the diagonal vectors.

The results obtained with both methods are compatible with each other, supporting the idea that input reduction is a *valid strategy* for scalable noise modeling. However, some differences emerge in the training behavior and final metrics: the diagonal-only approach shows higher variance in fidelity and trace distance during the early training stages, as reduced state information makes the agent's initial actions almost random, amplifying fluctuations across circuits. Over time, this gap narrows as the policy stabilizes.

Moreover, by comparing the two methods, it can be observed that the information most relevant for achieving high fidelity is largely concentrated on the diagonal, while trace distance pays the price of losing local information when off-diagonal elements are ignored. This explains why diagonal-only training achieves excellent global state overlap but displays increased noise in point-by-point distance.

3.5 Effect of Shot Statistics

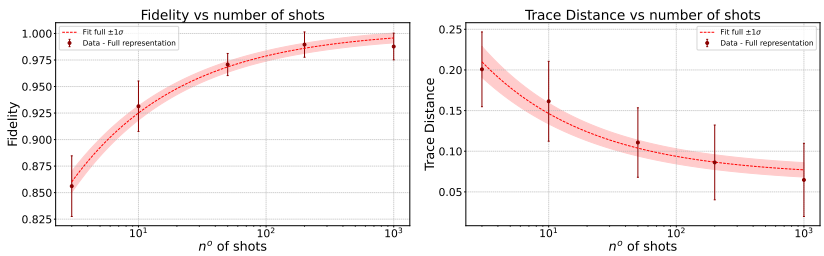
This experiment tests whether *input reduction* remains advantageous when the number of measurement shots is progressively lowered. A shot is a single execution of the quantum circuit followed by a full measurement of all qubits. The reduced representation is compared with the full density matrix for single-qubit and three-qubit circuits.

Starting from the ideal datasets introduced in Sec. 3.2, multiple shot budgets are considered. For each case, every circuit is first evolved through the noise model of Sec. 3.2 and then reconstructed by *state tomography*[7].

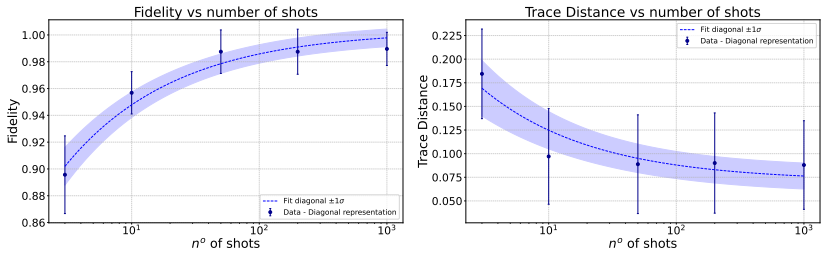
Since the tomographic reconstruction is based on estimating expectation values from repeated measurements, the statistical uncertainty on each observable scales as $1/\sqrt{n}$ where n is the number of shots. It is therefore expected the average fidelity to follow a saturation curve of the form:

$$\mathcal{F}(n) = \mathcal{F}_\infty - \frac{A}{\sqrt{n}}, \qquad TD(n) = TD_\infty + \frac{B}{\sqrt{n}}$$

A PPO agent is independently trained on every shot-dependent dataset. After convergence, each model is evaluated on the *ideal* test set (no shot noise), to reveal how well it generalizes beyond the statistical fluctuations seen during learning. The trend of both fidelity and trace distance (1 qubit) can be observed in Figure 3.3:



(a) 1 qubit - Full representation.

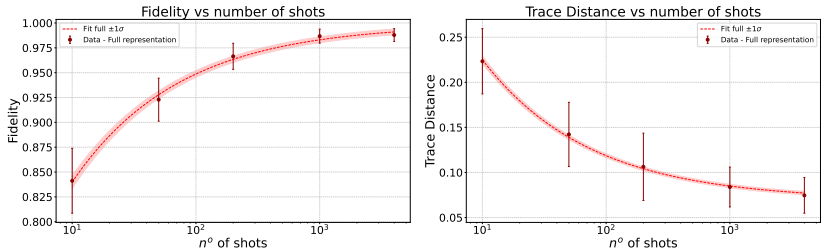


(b) 1 qubit - Diagonal representation.

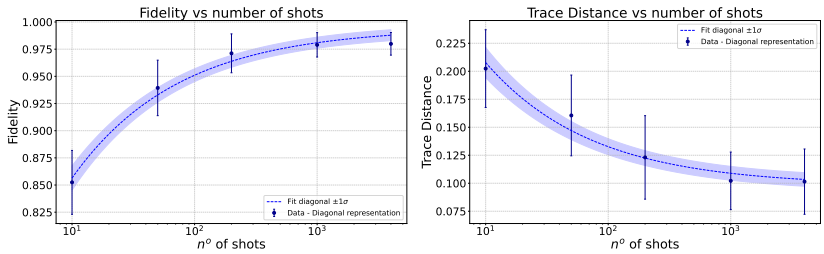
Figure 3.3. Fidelity and trace distance with respect to the number of shots for both the full and diagonal representations for 1 qubit. The curves highlight how performance degrades as measurements become scarce and clarify the relative robustness offered by the input-reduction strategy.

[7]Quantum state tomography reconstructs a state's density matrix ρ by repeatedly preparing identical copies, measuring an informationally complete set (e.g., Pauli X, Y, Z for qubits), and inverting the measurement model to obtain a physical, trace-one estimate.

Both fidelity and trace distance exhibit a clear inverse-square-root dependence on the number of shots. The diagonal-only dataset yields compatible fidelities to the full representation, indicating that the critical information is effectively captured along the diagonal. The results for three-qubit circuits are shown in Figure 3.4:



(a) 3 qubits - Full representation.



(b) 3 qubits - Diagonal representation.

Figure 3.4. Fidelity and trace distance with respect to the number of shots for both the full and diagonal representations for 3 qubits.

The parameters of the fit were estimated using `scipy.optimize`, and the asymptotic fidelities are compatible with $F_\infty = 1$ within 1σ for both the diagonal and full representations. In contrast, the asymptotic trace distances are not compatible with zero. Results are shown in Table 3.4

Qubits	Representation	A	F_∞	B	TD_∞
1	Full	0.25 ± 0.02	1.004 ± 0.005	0.24 ± 0.03	0.069 ± 0.009
	Diagonal	0.18 ± 0.02	1.003 ± 0.007	0.17 ± 0.05	0.071 ± 0.014
3	Full	0.50 ± 0.02	0.999 ± 0.003	0.49 ± 0.01	0.069 ± 0.002
	Diagonal	0.44 ± 0.03	0.995 ± 0.005	0.35 ± 0.04	0.098 ± 0.009

Table 3.4. Fit parameters estimation for all types of configurations.

The fidelity, being a global overlap measure, grows rapidly with the number of shots, while the trace distance remains bounded away from zero because fine-grained local information is lost—an effect slightly amplified when only the diagonal of the density matrix is used. Additionally, the number of shots required for the model to learn scales with system size: taking a fidelity of 0.95 as the threshold, the critical shot budget is ≈ 20 shots for the one-qubit *full* setting, ≈ 10 shots for the one-qubit *diagonal* setting, and ≈ 100 shots for both *full* and *diagonal* three-qubit scenarios.

Conclusions

The analysis conducted in this work has demonstrated that the *diagonal-only* approach is an effective method for input reduction in the quantum noise modeling problem. The performance obtained using only the diagonal elements of the density matrices was found to be comparable to the full data representation, both in the training phase and when varying the number of measurement shots.

This confirms that most of the relevant information for global performance metrics such as fidelity is concentrated in the diagonal elements, making this method a promising strategy for improving scalability.

The study of performance as a function of the number of shots further supports this conclusion: the asymptotic fidelities obtained from `scipy.optimize` fits are compatible with $F_\infty = 1$ for both full and diagonal cases, while the trace distance is not compatible with zero, confirming that local information cannot be perfectly reconstructed when using reduced inputs. Consistently with this picture, we observe a *critical shot budget* separating a stagnant regime from a learnable one: for one qubit, the threshold to reach a target fidelity $F \simeq 0.95$ is about 20 shots in the *full* setting and about 10 shots in the *diagonal* setting, whereas for three qubits it increases to roughly 100 shots in both the *full* and *diagonal* scenarios.

This scaling highlights how shot noise interacts with model capacity and input compression as system size grows.

These findings delineate both the promise and the limits of input reduction. On the one hand, diagonal-only inputs enable tangible scalability gains without sacrificing global fidelity; on the other, the non-vanishing trace distance cautions that purely local (point-wise) accuracy is curtailed.

An intermediate development could involve selecting *submatrices* to reconstruct the full density matrix. By focusing on these blocks, it becomes possible to retain the most relevant local quantum correlations while discarding less significant long-range terms, thereby reducing the dimensionality of the input.

For even larger systems (> 5 qubits), *graph neural networks* (GNNs) emerge as an attractive option. By naturally encoding the circuit connectivity and the spatial distribution of qubits, GNNs can explicitly exploit this topological information while avoiding the exponential blow-up of fully connected representations. Hybrid methods that combine reinforcement learning and GNNs [11] represent a particularly promising direction, enabling agents to exploit both the sequential decision-making framework of RL and the relational inductive biases of graph-based learning.

Overall, this work confirms the potential of input reduction as an effective strategy for quantum noise modeling, providing a foundation for future research toward more scalable frameworks.

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